

Four Frequencies Generator



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During development, testing and maintenance of digital and analogue electronic circuits and embedded systems, there is a need for TTL and CMOS signals produced by stable frequency sources. These sources are also useful for fast checking of oscilloscopes, probes, multimeters, frequency metres and other measurement equipment. This circuit produces fixed frequencies of 4MHz, 6MHz, 10MHz and adjustable square-wave with variable frequencies from around 10Hz to more than 100kHz.

Circuit and working

Fig. 1 shows the circuit of the four frequencies reference source. It is built with three 74C04 hex inverter ICs (IC1, IC2 and IC3) and one 74C14 hex inverter schmitt trigger IC (IC4). Each of the four blocks of the circuit in Fig. 1 produces a forward and inverted buffered signal.

IC1 uses crystal X_{TAL1} to produce the signal with a frequency of 10MHz. The produced output signal and inverted signal are available on connector CON1. Values of capacitors C1, C2 and C3 depend on the parameters of crystal X_{TAL1} and IC1.

IC2 uses crystal X_{TAL2} to produce the signal with a frequency of 6MHz. The produced output signal and inverted signal are available on connector CON2.

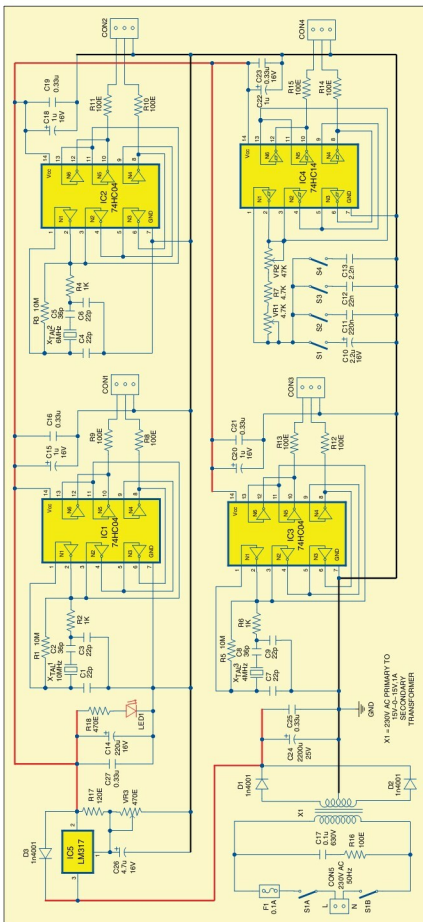


Fig. 1: Circuit diagram of the four frequencies generator